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ENGAGEMENT MECHANISM FOR VAPORIZER IN ANESTHESIA MACHINE

Abstract

An anesthesia machine includes an interface for receiving a vaporizer, a housing having an outer surface, a button having an outer surface, and configured to receive an engagement force from a user, and a latch including an engagement portion, and configured to selectively secure the vaporizer to the anesthesia machine via the engagement portion according to a position of the button, wherein a positional relationship between the outer surface of the button with the outer surface of the housing indicates whether or not the vaporizer is correctly received by the interface of the anesthesia machine

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] [Not Applicable]

BACKGROUND

[0002] Generally, this application relates to the engagement mechanism for a vaporizer when it is inserted and removed from an interface in an anesthesia machine.

SUMMARY

[0003] According to embodiments, an anesthesia machine for engaging a vaporizer with an interface of the anesthesia machine is provided, wherein the vaporizer includes an upper surface with a shelf and a recess, wherein the anesthesia machine includes: a button configured to receive an engagement force from a user and responsively move inwardly; a wedge including at least one upwardly-facing sloped region and at least one downwardly-facing sloped region; and a latch including an engagement portion configured to engage with the upper surface of the vaporizer, and including at least one laterally-projecting portion configured to engage with the at least one upwardly-facing sloped region and the at least one downwardly-facing sloped region of the wedge, wherein the engagement portion of the latch is configured to be forced upwardly when the vaporizer is being inserted into the interface of the anesthesia machine in response to the engagement portion of the latch engaging with the shelf of the upper surface of the vaporizer, and wherein the at least one laterally-projecting portion of the latch is configured to engage with the at least one downwardly-facing sloped region of the wedge when the engagement portion of the latch engages with the shelf of the upper surface of the vaporizer, thereby causing the wedge and the button to move inwardly, wherein the engagement portion of the latch is configured to be received by the recess of the upper surface of the vaporizer, thereby causing the wedge and the button to move outwardly, and wherein the button is configured to receive the engagement force while the engagement portion of the latch is received in the recess in the upper surface of the vaporizer, thereby causing the at least one upwardly-facing sloped region of the wedge to engage with the at least one laterally-projecting portion of the latch, thereby causing the engagement portion of the latch to be lifted upwardly out of the recess in the upper surface of the vaporizer. The anesthesia machine may further include a button return spring interposed between the button and a housing of the anesthesia machine, wherein the button return spring causes the button and the wedge to move outwardly. The anesthesia machine may further include a latch compression spring configured to exert a force on the latch thereby causing the engagement portion of the latch to exert a force downwardly. The at least one upwardly-facing sloped region of the wedge may include a first upwardly-facing sloped region and a second upwardly-facing sloped region; the at least one downwardly-facing sloped region of the wedge may include a first downwardly-facing sloped region and a second downwardly-facing sloped region; the at least one laterally-projecting portion of the latch may include a first laterally-projecting portion and a second laterally-projecting portion; wherein the first laterally-projecting portion of the latch may be configured to engage with the first upwardly-facing sloped region of the wedge and the first downwardly-facing sloped region of the wedge; and wherein the second laterally-projecting portion of the latch may be configured to engage with the second upwardly-facing sloped region of the wedge and the second downwardly-facing sloped region of the wedge. A slope of the at least one downwardly-facing sloped region may be greater than a slope of the at least one upwardly-facing sloped region. The vaporizer may include a vaporizer configured to contain a gas. The latch may include a pivot configured to be rotatably moved with respect to the housing, such that the latch is configured to rotate around the pivot. The wedge may include a hook including one of the at least one downwardly-facing sloped region and one of the at least one upwardly-facing sloped region. The wedge may include a plurality of rails configured to be received by a corresponding plurality of tracks in the housing of the anesthesia machine, and wherein the wedge further may include a plurality of spring arms configured to secure the wedge to the plurality of tracks. The interface of the anesthesia machine

may include a housing having an outer surface, wherein when an outer surface of the button is flush with the outer surface of the housing, it indicates that the vaporizer has been fully received by the interface, and wherein when the outer surface of the button is not flush with the outer surface of the housing, it indicates that the vaporizer has not been fully received by the interface.

[0004] According to embodiments, an anesthesia machine includes an interface for receiving a vaporizer, the anesthesia machine further includes: a housing having an outer surface; a button having an outer surface, and configured to receive an engagement force from a user; and a latch including an engagement portion, and configured to selectively secure the vaporizer to the anesthesia machine via the engagement portion according to a position of the button, wherein a positional relationship between the outer surface of the button with the outer surface of the housing indicates whether or not the vaporizer is correctly received by the interface of the anesthesia machine. When the outer surface of the button is not in a predetermined position with the outer surface of the housing, this may indicate that the vaporizer has not been fully received by the anesthesia machine. The predetermined position may be when the outer surface of the button is substantially flush with the outer surface of the housing. The anesthesia machine may further include a wedge configured to move with the button, wherein the wedge may be configured to interact with the latch to force the engagement portion of the latch away from the vaporizer when the button is forced inwardly. The wedge may include at least one upwardly-facing sloped region configured to interact with the latch to force the engagement portion of the latch away from the vaporizer. The wedge may be configured to interact with the latch and to pull the button inwardly when the engagement portion of the latch is being forced by the vaporizer. The wedge may include at least one downwardly-facing sloped region configured to interact with the latch to cause the wedge to pull the button inwardly when the engagement portion of the latch is being forced by the vaporizer.

[0005] According to embodiments, a method for inserting a vaporizer into an interface of an anesthesia machine includes: partially receiving the vaporizer in the interface, thereby causing a button of the anesthesia machine to not be in a predetermined position; and fully receiving the vaporizer in the interface, thereby causing the button of the anesthesia machine to be in a predetermined position. When the vaporizer is fully received by the interface, the vaporizer may be engaged with the anesthesia machine. When the button receives a force from a user, the vaporizer may be disengaged with the anesthesia machine.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

[0006] FIG. 1 illustrates an anesthesia machine and two removable vaporizers, according to embodiments.

[0007] FIG. 2A illustrates a portion of an anesthesia machine and two removable vaporizers, with the vaporizers fully inserted into the anesthesia machine, according to embodiments.

[0008] FIG. 2B illustrates a portion of an anesthesia machine and one removable vaporizer, where the anesthesia machine has an empty interface for receiving another removable vaporizer, according to embodiments.

[0009] FIG. 2C illustrates a portion of an anesthesia machine and two removable vaporizers, where one of the removable vaporizers is not fully inserted into an interface of the anesthesia machine, according to embodiments.

[0010] FIG. 3A illustrates a top perspective view of the engagement mechanism for engaging a removable vaporizer in an interface of an anesthesia machine, according to embodiments.

[0011] FIG. 3B illustrates a bottom perspective view of the engagement mechanism for engaging a removable vaporizer in an interface of an anesthesia machine, according to embodiments.

[0012] FIG. 3C illustrates an exploded view of the engagement mechanism for engaging a removable vaporizer in an interface of an anesthesia machine, according to embodiments.

[0013] FIG. 3D illustrates an exploded view of a portion of the engagement mechanism for engaging a removable vaporizer in an interface of an anesthesia machine, according to embodiments.

[0014] FIG. 4A illustrates a cross-sectional view of a portion of an anesthesia machine including the engagement mechanism, and a portion of a removable vaporizer, where the removable vaporizer is fully inserted into an interface of the anesthesia machine, according to embodiments.

[0015] FIG. 4B illustrates a cross-sectional view of a portion of an anesthesia machine including the engagement mechanism, and a portion of a removable vaporizer, where the removable vaporizer is partially inserted into an interface of the anesthesia machine, according to embodiments.

[0016] FIG. 4C illustrates a cross-sectional view of a portion of an anesthesia machine including the engagement mechanism, and a portion of a removable vaporizer, where the removable vaporizer is fully inserted into an interface of the anesthesia machine, and where the engagement mechanism has been engaged by a user to disengage the engagement mechanism from the removable vaporizer, according to embodiments.

[0017] FIG. 5A illustrates a cross-sectional view with perspective of a portion of an anesthesia machine including the engagement mechanism, and a portion of a removable vaporizer, where the removable vaporizer is fully inserted into an interface of the anesthesia machine, according to embodiments.

[0018] FIG. 5B illustrates a cross-sectional view with perspective of a portion of an anesthesia machine including a portion of the engagement mechanism, according to embodiments.

[0019] FIG. 6 is a flowchart of a method for engaging and disengaging a vaporizer from an anesthesia machine, according to embodiments.

[0020] The foregoing summary, as well as the following detailed description of certain techniques of the present application, will be better understood when read in conjunction with the appended drawings. For the purposes of illustration, certain techniques are shown in the drawings. It should be understood, however, that the claims are not limited to the arrangements and instrumentality shown in the attached drawings. Furthermore, the appearance shown in the drawings is one of many ornamental appearances that can be employed to achieve the stated functions of the system.

DETAILED DESCRIPTION

[0021] Embodiments of an anesthesia machine are disclosed herein that receives a removable vaporizer. The anesthesia machine includes an engagement mechanism, embodiments of which are described herein. The engagement mechanism may lock the vaporizer automatically once the vaporizer is inserted, and the engagement mechanism may unlock the vaporizer once a button is pressed. The engagement mechanism may further provide a safety/warning indicator as well, in case the vaporizer is unlatched or not fully received at the interface of the anesthesia machine. This indication may occur even when no power is supplied to the anesthesia machine and/or vaporizer, as the engagement mechanism may be implemented mechanically only. The button of the engagement mechanism may have a predetermined position when the vaporizer is fully inserted into the interface of the anesthesia machine. If the button is not in the predetermined position, then that may indicate that the vaporizer is not fully inserted.

[0022] According to embodiments, when the vaporizer is inserted into the interface of the anesthesia machine, a latch in the engagement mechanism may be lifted and then may drop into a recess on the outer surface of the vaporizer. When the latch is lifted up, the button may not be in the predetermined position. When the latch drops into the recess on the outer surface of the vaporizer, the button may be placed in the predetermined position. Further, when the button is pressed, the latch may be lifted out of the recess on the outer surface of the vaporizer, thereby disengaging the vaporizer and allowing for its removal. The button may return to the predetermined position after

vaporizer is removed from the anesthesia machine by the user.

[0023] According to embodiments, the engagement mechanism includes a wedge having sloped regions, a latch, and a button. The wedge may interact with the latch and the button. Certain sloped region(s) in the wedge cause the latch to be removed from a recess in the vaporizer when a user presses the button. Removing the latch from the recess may disengage the vaporizer from the anesthesia machine, such that a user may remove the vaporizer. When the latch is forced away from the vaporizer, this may cause the wedge and the button to move away from a predetermined position. When the button is not in the predetermined position, this may serve as an alert that the vaporizer has not been fully received by the interface of the anesthesia machine.

[0024] While embodiments described herein primarily relate to an anesthesia machine and a vaporizer, the engagement mechanism may be incorporated into other types of devices or systems other than an anesthesia machine. Further, instead of a vaporizer, the engagement mechanism may selectively engage or disengage another kind of mechanical component. Thus, the disclosure is not limited to the specific combination of anesthesia machines and vaporizers.

[0025] Aspects of the present disclosure have the technical effect of engaging and disengaging a vaporizer from an anesthesia machine. Various embodiments have the technical effect of indicating to a user that the vaporizer is not fully received by an interface of the anesthesia machine. Various embodiments have the technical effect of providing the disclosed functionality without the need for electric power.

[0026] FIG. 1 illustrates an anesthesia machine **10** and two removable vaporizers **100**, according to embodiments. The anesthesia machine **10** is used by a clinician to deliver anesthetic gas to a patient. Exemplary embodiments of the anesthesia machine **10** are disclosed in U.S. Pat. Nos. 11,318,274 and 11,439,789, which are incorporated by reference in their entireties herein. Exemplary vaporizers **100** may be devices that induce and control vaporization of a volatile anesthetic agent from a liquid form. Exemplary embodiments of a vaporizer **100** are disclosed in U.S. Pat. Nos. 11,318,274 and 11,439,789, which are incorporated by reference in their entireties herein.

[0027] FIG. 2A illustrates a portion of the anesthesia machine **10** and two removable vaporizers **100**, with the vaporizers **100** fully received by corresponding interfaces **11** of the anesthesia machine **10**, according to embodiments. FIG. 2A further shows two buttons **210** from two corresponding engagement mechanisms **200** (not fully shown). The buttons **210** are configured to receive a disengagement force from a user to disengage the corresponding vaporizers **100** from the anesthesia machine **10**, such that the corresponding vaporizers **100** can be removed. The interfaces **11** of the anesthesia machine **10** are not visible in FIG. 2A, because the vaporizers **100** are fully received. The buttons **210** are in a predetermined position indicating that the corresponding vaporizers **100** are fully received or engaged with the anesthesia machine **10**. In the depicted embodiment, the predetermined position is when an outer surface of a given button **210** is substantially flush with the housing **14** (e.g., surrounding housing) of the anesthesia machine **10**.

[0028] FIG. 2B illustrates a portion of the anesthesia machine **10** and a removable vaporizer **100** fully received into a corresponding interface **11** (not visible) of the anesthesia machine **10**, and further illustrating an empty interface **11** configured to receive another vaporizer **100** (not shown), according to embodiments. FIG. 2B further shows the two buttons **210** from the two corresponding engagement mechanisms **200** (not fully shown). The left-side button **210** is in a predetermined position indicating that the corresponding vaporizer **100** is fully received or engaged with the anesthesia machine **10**. In the depicted embodiment, the predetermined position is when the outer surface of the left-side button **210** is substantially flush with the housing **14** (e.g., surrounding housing) of the anesthesia machine **10**. In the depicted embodiment, the right-side button **210** is also in a predetermined position when a corresponding vaporizer **100** has not been inserted into the interface **11** at all. In the depicted embodiment, the predetermined position for a given button **210** when a corresponding vaporizer **100** has not been inserted into the interface **11** at all is that the

outer surface of the given button **210** is substantially flush with the housing **14** (e.g., surrounding housing) of the anesthesia machine **10**.

[0029] FIG. 2C illustrates a portion of the anesthesia machine **10** and two removable vaporizers **100**, with the left-side vaporizer **100** fully received by the left-side interface **11** of the anesthesia machine **10**. This arrangement is similar to what is shown with the left-side vaporizer **100** in FIGS. 2A and 2B. Correspondingly, the left-side button **210** is in the predetermined position. FIG. 2C further shows the right-side vaporizer **100** being only partially received by the right-side interface **11** of the anesthesia machine **10**. In this embodiment, the outer surface of the button **210** is not in the predetermined position (e.g., it is recessed from the housing **14**, e.g., surrounding housing, of the anesthesia machine **10**).

[0030] FIGS. 3A, 3B, 3C, and 3D illustrate different views of the engagement mechanism **200** for engaging the vaporizer **100** in the interface **11** of the anesthesia machine **10**, according to embodiments. As shown in FIGS. 1, 2A, 2B, and 2C, the anesthesia machine **10** includes or houses two such engagement mechanisms **200**, although one, three, or more engagement mechanisms **200** may be included or housed. Particularly, FIG. 3A illustrates a top perspective view of the engagement mechanism **200** for engaging the vaporizer **100** (not shown) in the interface **11** (not shown) of the anesthesia machine **10** (not shown), according to embodiments. FIG. 3B illustrates a bottom perspective view of the engagement mechanism **200** shown in FIG. 3A, according to embodiments. FIG. 3C illustrates an exploded view of the engagement mechanism **200** shown in FIGS. 3A and 3B, according to embodiments. FIG. 3D illustrates an exploded view of a portion of the engagement mechanism **200** shown in FIGS. 3A, 3B, and 3C, according to embodiments.

[0031] In the overall embodiment illustrated, engagement mechanism **200** includes a button **210**, a button-return spring **220**, a sheath **230**, a latch **240**, a wedge **250**, a latch compression spring **260**, fasteners **270**, a nipple **280**, and inserts **290**, although more or fewer components may be used according to embodiments.

[0032] The button **210** includes an outer surface **213** configured to allow interaction with a user (e.g., interaction with a user's hand or fingers). The button **210** may be an integral piece, as shown, or may be formed by a plurality of coupled portions, such as the different portions described herein. The button **210** may include a shaft **211** and one or more insert-receiving holes **212**. The shaft **211** may be cylindrical. The shaft **211** may receive the button-return spring sleeve **230** (e.g., coaxially). The shaft **211** may further receive the button-return spring **220** (e.g., coaxially). The insert-receiving hole(s) **212** may receive the inserts **290**.

[0033] The button-return spring **220** may be a coil spring or other type of spring. The button-return spring **220** may be formed or include metal or plastic. The button-return spring **220** may be received by the button shaft **211** and/or the sleeve **230**. The button-return spring **220** may abut the button **210** and the sleeve (e.g., a collar of the sleeve **230**). The button-return spring **220** may be interposed between the button **210** and the sleeve **230** (e.g., a collar **231** of the sleeve **230**) and housing **14** of the anesthesia machine **10** (not shown in FIGS. 3A-3D). The button-return spring **220** may be coupled to the button **210** and/or the sleeve **230** (e.g., the collar **231** of the sleeve **230**). Instead of the sleeve **230**, the button-return spring **220** may couple directly with the housing **14** of the anesthesia machine **10**. The button-return spring **220** may couple with the button **210**, sleeve **230**, and/or housing **14** of the anesthesia machine **10** by virtue of compression of the button-return spring **220** (e.g., a given end of the button-return spring **220** may exert a force against another component).

[0034] The sleeve **230** may include a collar **231** and insert **232**. The sleeve **230** may be received by or receive the shaft **211** of the button **210**. The sleeve **230** may further be received by or receive a portion of the housing **14** of the anesthesia device **10** (e.g., through an aperture in the housing **14**). The sleeve **230** may otherwise be coupled to the housing **14**. The sleeve **230** may engage with the housing **14** via a sleeve insert **232**. The sleeve **230** may not change position substantially during operation of the engagement mechanism **200**. The sleeve **230** (and/or the housing **14**) may limit an

inward movement of the button **210**.

[0035] The nipple **280** may include an insert **281** and a pad **282**. The nipple **280** may be an integral piece or a collection of smaller portions coupled together, such as the portions described herein. The insert **281** may be inserted into an aperture in the wedge **250** to secure the nipple **280** to the wedge **250**. The pad **282** may extend from the insert **281** and face towards the sheath **230**.

[0036] The latch **240** may include one or more laterally projecting portions **241**, a pivot **242**, an engagement portion **243**, and a latch spring aperture **244**, as shown. The latch **240** may be one integral piece or be formed from a collection of different portions, such as the specific portions described herein. The latch **240** may include a lattice as shown to reduce the weight and expense of the latch **240**. The laterally projecting portion(s) **241** may be round (e.g., circular or cylindrical). The laterally projecting portion(s) **241** may have a hollow interior region. The laterally projecting portion(s) **241** may engage or interact with the wedge **250**. The pivot **242** may interface with the housing **14** of the anesthesia machine **10** (either by abutting the housing **14** or indirectly interfacing with the housing). The latch **240** may rotate about the pivot **242**. The engagement portion **243** may project downwardly as shown, or may project in another direction. The engagement portion **243** may selectively engage with a vaporizer **100**. The engagement portion **243** may include a substantially rounded side and a substantially vertical side. The latch spring aperture **244** may receive the latch spring **260**, such that a portion of the latch spring **260** is contained within the latch **240**.

[0037] The wedge **250** may include one or more hook(s) **251**, one or more upwardly-facing sloped region(s) **252**, one or more downwardly-facing sloped region(s) **253**, one or more fastener-receiving apertures **254**, and one or more wedge housing engagement portions **255**, according to the embodiment depicted. The wedge **250** may be an integral piece, as shown, or may be formed by a plurality of coupled portions, such as the different portions described herein. There may be two hooks **251**, one on the right side and one on the left. The hook(s) **251** may be elongated and extend along the dimension along which the wedge **250** travels. Each hook **251** may have an upwardly-facing sloped region **252** and a downwardly-facing sloped region **253**. The upwardly-facing sloped region **252** and the downwardly-facing sloped region **253** may extend towards each other into a U-shaped region or V-shaped region that connects the two. The fastener-receiving aperture(s) **254** may receive corresponding fastener(s) **270**. The wedge housing engagement portion(s) **255** may engage with the housing **14** (not shown) to slide inwardly and outwardly along the housing **14** while being constrained from movement in other directions. The wedge housing engagement portion(s) **255** may include rails as shown, which are received by tracks in the housing **14** (not shown). The wedge housing engagement portion(s) **255** may include spring arms as shown, which engage with the housing **14** to stabilize the wedge **250** as it travels inwardly and outwardly in the anesthesia machine **10**. The wedge housing engagement portion(s) **255** may further have lateral recess(es) to reduce the amount of surface area on the wedge housing engagement portion(s) **255** that contacts the housing **14**.

[0038] The latch compression spring **260** may be a coil spring or other type of spring. The latch compression spring **260** may be formed or include metal or plastic. The latch compression spring **260** may engage or couple with the housing **14** on one end (e.g., directly or indirectly engage or couple with the housing **14**). The latch compression spring **260** on the other side may extend through the latch spring aperture **244** in the latch **240** and may engage or couple with the latch **240** (e.g., directly or indirectly engage or couple the latch **240**). The latch compression spring **260** may engage or couple with the housing **14** and latch **240** by virtue of compression of the latch compression spring **260** (e.g., a given end of the latch compression spring **260** may exert a force against another component).

[0039] The fastener(s) **270** may be bolts or screws or other types of fasteners. The fastener(s) **270** may couple (e.g., statically couple) the button **210** and the wedge **250**. The fastener(s) may extend through the fastener-receiving aperture(s) **254** in the wedge **250** and be received by the button hole

insert(s) **290** located in the insert-receiving hole(s) **212** in the button **210**.

[0040] The engagement mechanism **200** may be configured as follows. The shaft **211** of the button **210** may receive a hollow interior region of the sleeve **230**. The button-return spring **220** may extend over the shaft **211** of the button **210** and the sleeve **230**. The button-return spring **220** may always remain compressed to a degree such that it always exerts forces against the button **210** and the sleeve **230** (e.g., the collar **231** of the sleeve **230**). The insert **232** of the sleeve **230** may be inserted into an aperture in the housing **14** of the anesthesia machine **10**. The inserts **280** may be received by the insert-receiving apertures **212** in the button **210**. The fasteners **270** may extend through the faster-receiving apertures **254** and received by the inserts **280** to couple (e.g., statically couple) the button **210** with the wedge **250**. The nipple **280** may engage with the wedge **250** via the nipple insert **281** such that the nipple **280** moves with the wedge **250**. The laterally-extending portions **241** of the latch **240** may be positioned in or on corresponding hooks **251** in the wedge **250**. The wedge **250** may be moveable inwardly and outwardly along the housing **14** via the wedge housing engagement portions **255** and corresponding features on the housing **14**. The pivot **242** of the latch **240** may press against the housing **14**, such that the latch **240** rotates around the point where the pivot **242** presses against the housing **14**. The latch compression spring **260** may engage with the latch **260** through the latch spring aperture **244** and with the housing **14**.

[0041] FIGS. **4A**, **4B**, and **4C** show different modes of operation of the engagement mechanism **200** with respect to the housing **14** of the anesthesia machine **10** and the vaporizer **100**, according to embodiments. FIG. **4A** illustrates a cross-sectional view of a portion of an anesthesia machine **10** including the engagement mechanism **200**, and a portion of a removable vaporizer **100**, where the vaporizer **100** is fully inserted into an interface **11** of the anesthesia machine **10**, according to embodiments. FIG. **4B** illustrates a cross-sectional view of a portion of the anesthesia machine **10** including the engagement mechanism **200**, and a portion of the vaporizer **100**, where the vaporizer **100** is partially inserted into the interface **11** of the anesthesia machine **10**, according to embodiments. FIG. **4C** illustrates a cross-sectional view of a portion of the anesthesia machine **10** including the engagement mechanism **200**, and a portion of the vaporizer **100**, where the vaporizer **100** is fully inserted into the interface **11** of the anesthesia machine **10**, and where a user has interacted with the engagement mechanism **200** to disengage the engagement mechanism **200** from the vaporizer **100**, according to embodiments. FIGS. **4A**, **4B**, and **4C** show only one hook **251** in the wedge **250**, but it is understood that a similar arrangement may be optionally included for another hook **251**, as depicted in FIGS. **3A-3D**.

[0042] As shown in FIG. **4A**, the vaporizer **100** may include a wall having an outer surface. The outer surface forms a recess **102** and a shelf **104**. When the vaporizer **100** has been fully received by the interface **11** of the anesthesia machine **10**, the engagement portion **243** of the latch **240** may extend into the recess **102** of the vaporizer **100**. A force may be exerted by the latch compression spring **260** such that the engagement portion **243** presses downwardly into the recess against the vaporizer. If there is no force exerted, the engagement portion **243** of the latch **240** may still extend into the recess **102** of the vaporizer **100**. The shape of the engagement portion **243** of the latch **240** may assist in securing the vaporizer **100** in the interface **11** of the anesthesia machine **10**. As shown, the engagement portion **243** of the latch may only abut the recess **102** of the vaporizer **100** at a limited surface area. If a pulling force is exerted on the vaporizer **100**, the shape of the recess **102** and the shape of the engagement portion **243** may prevent the engagement portion **243** from moving upwardly, in addition to the effect of a downward force being applied to the engagement portion **243**.

[0043] As shown in FIG. **4A**, the latch compression spring **260** may be compressed, but less compressed than in FIGS. **4B** and **4C**. The latch **240** may be rotated counterclockwise around the location where the pivot **242** abuts the housing **14**. The laterally extending portion **241** of the latch **240** may extend into corresponding U-shaped region in the hook **251**. The laterally extending portion **241** may be fully received by the hook **251** of the wedge **250**. The position of the laterally

extending portion **241** with respect to the hook **251** may determine the location of the wedge **250**. When the button **210** is coupled to the wedge **250** (e.g., statically coupled), the position of the button **210** may further be determined with respect to the position of the wedge **250**. In the arrangement of FIG. 4A, the outer surface **213** of the button **210** may be at a predetermined location. In this embodiment, the predetermined location of the outer surface **213** of the button **210** may be substantially flush with respect to the outer surface of the housing **14** of the anesthesia machine **10** proximate the button **210**. When the outer surface **213** of the button **210** is in the predetermined position, this indicates that the vaporizer **100** may be fully received by the interface **11** of the anesthesia machine **10** and/or that the engagement mechanism **200** may be engaged the vaporizer **100**. The button return spring **220** may be uncompressed to its maximum extent, thereby maintaining the outer surface **213** of the button **210** in the predetermined position.

[0044] As shown in FIG. 4B, when the vaporizer **100** is not fully received by the interface **11** of the anesthesia machine **10**, the engagement portion **243** of the latch **240** may be forced upwardly by the shelf **104** of the vaporizer **100**. This may cause the latch **240** to rotate clockwise around the point where the pivot **242** abuts the housing **14**. This may further cause the latch compression spring **260** to compress further, and may cause the laterally extending portion **241** of the latch **240** to engage with the downwardly-facing sloped region **253** of the wedge **250**. As the laterally extending portion **241** presses against the downwardly-facing sloped region **253** of the wedge **250**, this may force the wedge **250** to move inwardly, thereby causing the button **210** to be pulled inwardly. When the button **210** is pulled inwardly, the outer surface **213** of the button **210** may be moved out of the predetermined position. This may indicate that the vaporizer **100** has not been fully received by the interface **11** of the anesthesia machine **10**. This may further compress the button-return spring **220**, causing relatively stronger forces to be exerted against the button **210** and the housing **14** via the sleeve **230**.

[0045] As shown in FIG. 4C, when the outer surface **213** of the button **210** is engaged (e.g., by a user's finger or hand) and forced inwardly, this may cause the button **210** to move inwardly, as well as the wedge **250** to move inwardly as well. This inward movement may also compress the button-return spring **220**, such that the button **210** may be forced outwardly once the engagement force is removed from the outer surface **213** of the button **210**. As the wedge **250** is moved inwardly, the laterally projecting portion **241** of the latch **240** may engage with the upwardly-facing sloped region **252** on the hook **251** of the wedge **250**. As the wedge **250** moves farther inwardly, the latch **240** may rotate clockwise around the location where the pivot **242** abuts the housing **14**. This may cause the latch engagement portion **243** of the latch **240** to be repositioned such that it is no longer within the recess **102** of the vaporizer **100**. Once the latch engagement portion **243** of the latch **240** has been removed from the recess **102** of the vaporizer **100**, the vaporizer **100** may be removed from the interface **11** of the anesthesia machine **10**.

[0046] FIG. 5A is similar to FIG. 4B, except that the system is shown with perspective to provide additional context about the arrangement of the anesthesia machine **10**, the engagement mechanism **200**, and the vaporizer **100**. FIG. 5B illustrates a cross-sectional view along the plane of the right-side fastener **270** to show the arrangement of the button **210**, the wedge **250**, the housing **14**, the spring **220**, and the nipple **280**.

[0047] FIG. 6 shows a flowchart **600** of a method for engaging and disengaging a vaporizer **100** from an anesthesia machine **10**, according to embodiments. The method is described in context of vaporizer **100** and anesthesia machine **10** (including the engagement mechanism **200** and the button **210**), but the method is not so limited. At step **610**, the vaporizer **100** is partially received at the interface **11** of the anesthesia machine **10**. This causes the button **210** of the engagement mechanism **200** of the anesthesia machine **10** to not be in a predetermined position (e.g., a predetermined position with respect to the housing **14** of the anesthesia machine **10**). FIG. 4B is an illustrative embodiment. At step **620**, the vaporizer **100** is fully received by the interface **11** of the anesthesia machine **10**. This causes the button **210** of the engagement mechanism of the anesthesia

machine **10** to be in the predetermined position. FIG. 4A is an illustrative embodiment. At step **630**, when the vaporizer **100** is fully received by the interface **11** of the anesthesia machine **10**, the vaporizer **100** is engaged by the anesthesia machine **10**. The vaporizer **100** is engaged such that it cannot be non-destructively removed from the anesthesia machine **10** or otherwise moved out of place in the interface **11** of the anesthesia machine **10**. FIG. 4A is an illustrative embodiment. At step **640**, the button **210** receives a force from a user (e.g., a force applied to the outer surface **213** of the button **210**), thereby causing the vaporizer **100** to be disengaged from the anesthesia machine **10**, such that the vaporizer **100** can be removed (e.g., removed by the user) from the anesthesia machine **10** or otherwise moved within the interface **11** of the anesthesia machine **10**. FIG. 4C is an illustrative embodiment.

TABLE-US-00001 Parts List Part Name Reference Anesthesia machine 10 Anesthesia machine interface 11 Anesthesia machine housing 14 Vaporizer 100 Vaporizer upper surface recess 102 Vaporizer upper surface shelf 104 Engagement mechanism 200 Button 210 Button shaft 211 Button insert-receiving apertures 212 Button outer surface 213 Button-return spring 220 Sleeve 230 Sleeve collar 231 Sleeve insert 232 Latch 240 Latch lateral portions 241 Latch pivot 242 Latch engagement portion 243 Latch spring aperture 244 Wedge 250 Wedge hooks 251 Wedge upwardly-facing sloped regions 252 Wedge downwardly-facing sloped regions 253 Wedge fastener-receiving apertures 254 Wedge housing engagement portions 255 Latch compression spring 260 Fasteners 270 Nipple 280 Nipple insert 281 Nipple pad 282 Inserts 290 Flow chart 300

[0048] It will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the novel techniques disclosed in this application. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the novel techniques without departing from its scope. Therefore, it is intended that the novel techniques not be limited to the particular techniques disclosed, but that they will include all techniques falling within the scope of the appended claims.

Claims

1. An anesthesia machine for engaging a vaporizer with an interface of the anesthesia machine, wherein the vaporizer includes an upper surface with a shelf and an recess, the anesthesia machine comprising: a button configured to receive an engagement force from a user and responsively move inwardly; a wedge including at least one upwardly-facing sloped region and at least one downwardly-facing sloped region; and a latch including an engagement portion configured to engage with the upper surface of the vaporizer, and including at least one laterally-projecting portion configured to engage with the at least one upwardly-facing sloped region and the at least one downwardly-facing sloped region of the wedge, wherein the engagement portion of the latch is configured to be forced upwardly when the vaporizer is being inserted into the interface of the anesthesia machine in response to the engagement portion of the latch engaging with the shelf of the upper surface of the vaporizer, and wherein the at least one laterally-projecting portion of the latch is configured to engage with the at least one downwardly-facing sloped region of the wedge when the engagement portion of the latch engages with the shelf of the upper surface of the vaporizer, thereby causing the wedge and the button to move inwardly, wherein the engagement portion of the latch is configured to be received by the recess of the upper surface of the vaporizer, thereby causing the wedge and the button to move outwardly, and wherein the button is configured to receive the engagement force while the engagement portion of the latch is received in the recess in the upper surface of the vaporizer, thereby causing the at least one upwardly-facing sloped region of the wedge to engage with the at least one laterally-projecting portion of the latch, thereby causing the engagement portion of the latch to be lifted upwardly out of the recess in the upper surface of the vaporizer.

2. The anesthesia machine of claim 1, further comprising a button return spring interposed between

the button and a housing of the anesthesia machine, wherein the button return spring causes the button and the wedge to move outwardly.

3. The anesthesia machine of claim 1, further comprising a latch compression spring configured to exert a force on the latch thereby causing the engagement portion of the latch to exert a force downwardly.

4. The anesthesia machine of claim 1, wherein: the at least one upwardly-facing sloped region of the wedge includes a first upwardly-facing sloped region and a second upwardly-facing sloped region; the at least one downwardly-facing sloped region of the wedge includes a first downwardly-facing sloped region and a second downwardly-facing sloped region; the at least one laterally-projecting portion of the latch includes a first laterally-projecting portion and a second laterally-projecting portion; wherein the first laterally-projecting portion of the latch is configured to engage with the first upwardly-facing sloped region of the wedge and the first downwardly-facing sloped region of the wedge; and wherein the second laterally-projecting portion of the latch is configured to engage with the second upwardly-facing sloped region of the wedge and the second downwardly-facing sloped region of the wedge.

5. The anesthesia machine of claim 1, wherein a slope of the at least one downwardly-facing sloped region is greater than a slope of the at least one upwardly-facing sloped region.

6. The anesthesia machine of claim 1, wherein the vaporizer comprises a vaporizer configured to contain a gas.

7. The anesthesia machine of claim 1, wherein the latch comprises a pivot configured to be rotatably moved with respect to the housing, such that the latch is configured to rotate around the pivot.

8. The anesthesia machine of claim 1, wherein the wedge includes a hook including one of the at least one downwardly-facing sloped region and one of the at least one upwardly-facing sloped region.

9. The anesthesia machine of claim 1, wherein the wedge comprises a plurality of rails configured to be received by a corresponding plurality of tracks in the housing of the anesthesia machine, and wherein the wedge further comprises a plurality of spring arms configured to secure the wedge to the plurality of tracks.

10. The anesthesia machine of claim 1, wherein the interface of the anesthesia machine comprises a housing having an outer surface, wherein when an outer surface of the button is flush with the outer surface of the housing, it indicates that the vaporizer has been fully received by the interface, and wherein when the outer surface of the button is not flush with the outer surface of the housing, it indicates that the vaporizer has not been fully received by the interface.

11. A system including an interface configured to retain and selectively eject a mechanical component, the system comprising: a housing having an outer surface; a button having an outer surface, and configured to receive an engagement force from a user; and a latch including an engagement portion, and configured to selectively secure the mechanical component to the system via the engagement portion according to a position of the button, wherein a positional relationship between the outer surface of the button with the outer surface of the housing indicates whether or not the mechanical component is correctly received by the interface of the system.

12. The system of claim 11, wherein when the outer surface of the button is not in a predetermined position with the outer surface of the housing, this indicates that the mechanical component has not been fully received by the system.

13. The system of claim 12, wherein the predetermined position is when the outer surface of the button is substantially flush with the outer surface of the housing.

14. The system of claim 11, further comprising a wedge configured to move with the button, wherein the wedge is configured to interact with the latch to force the engagement portion of the latch away from the mechanical component when the button is forced inwardly.

15. The system of claim 14, wherein the wedge comprises at least one upwardly-facing sloped

region configured to interact with the latch to force the engagement portion of the latch away from the mechanical component.

16. The system of claim 14, wherein the wedge is configured to interact with the latch and to pull the button inwardly when the engagement portion of the latch is being forced by the mechanical component.

17. The system of claim 16, wherein the wedge comprises at least one downwardly-facing sloped region configured to interact with the latch to cause the wedge to pull the button inwardly when the engagement portion of the latch is being forced by the mechanical component.

18. A method for inserting a vaporizer into an interface of an anesthesia machine, comprising: partially receiving the vaporizer in the interface, thereby causing a button of the anesthesia machine to not be in a predetermined position; and fully receiving the vaporizer in the interface, thereby causing the button of the anesthesia machine to be in a predetermined position.

19. The method of claim 18, wherein when the vaporizer is fully received by the interface, the vaporizer is engaged with the anesthesia machine.

20. The method of claim 19, wherein when the button receives a force from a user, the vaporizer is disengaged with the anesthesia machine.
